



Principal, Data Architect

[Apply](#) Sunnyvale, CA Full time Posted 3 Days Ago R-2404723

[Position Summary...](#)

[What you'll do...](#)

The Mission

As a Principal Data Architect, you are the visionary designer of Sam's Club's **Cognitive Data Infrastructure**. Your goal is to move beyond static data warehousing to build a **Semantic World Model**—a unified, real time, and context aware layer that serves as the "source of truth" for intelligent agents. You will architect the frameworks that allow AI to not only retrieve data but to reason across it, maintain persistent memory, and execute complex business logic with precision.

What You'll Do

- **Architect the Semantic World Model:** Lead the design of a global semantic layer that unifies structured, semi structured, and unstructured data into shared, context aware representations (Knowledge Graphs and Metadata Tensors) optimized for LLM reasoning.
- **Design for Real Time State:** Define the architecture for persistent agent memory and real time state

management. Ensure agents have a seamless "mental model" that synchronizes a member's current session with their long term history across all channels.

- **Establish the Tool-Calling Fabric:** Transition the enterprise from "Data-as-a-Table" to "**Data-as-a-Tool** for Agentic Enablement" Design the architectural contracts and schemas that allow agents to call data products as executable functions with deterministic outcomes.
- **Lead the Agentic Reference Architecture:** Define and steward enterprise standards for **Agent Ready Data**. This includes creating reference patterns for GraphRAG (Graph-Augmented Generation), vector space optimization, and hierarchical ontology design.
- **Enable Model-Agnostic Intelligence:** Build a decoupled semantic interface that ensures our data strategy remains stable and performant regardless of the underlying LLM or agent framework being utilized.
- **Architect Trust & Grounding:** Design "Reasoning Guardrails" into the data layer. Ensure that the semantic architecture provides verifiable grounding for AI actions, minimizing hallucinations through strict policy enforcement and deterministic data paths.
- **Strategic Influence & Modernization:** Partner with Engineering, AI/ML, and Product leaders to drive an **AI-native operating model**. Lead the technical roadmap for modernizing legacy systems into composable, intelligent platforms.

What You'll Bring

- **10–15+ years of experience** in large scale distributed data architecture, with a proven track record of shaping enterprise level data strategies.
- **Architectural Fluency in Agentic Systems:** Deep expertise in the "Context Stack"—including Vector Databases, Enterprise Knowledge Graphs, and the orchestration of multi agent workflows.
- **Semantic Mastery:** Expert level knowledge of **Ontology Engineering** and **Metadata Management**. You understand how to mathematically represent business logic so it is computable by a model.
- **Real Time Architecture Expertise:** Deep understanding of real time event processing and how to blend high volume streaming data with analytical stores to provide agents with "Total Recall."
- **Systems Thinking:** Ability to reason about the trade-offs between latency, information density, and token cost when designing data structures for LLM context windows.
- **Engineering Foundation:** Strong hands on experience across the modern data and AI stack:
 - **Core & Distributed Systems:** Python, Java, or Scala; **Databricks, Spark, and BigQuery.**
 - **Real Time & Streaming:** Kafka, Druid, and streaming frameworks like **Spark Structured**

Streaming, Kafka Connect, or Apache Flink.

o

Agentic Frameworks & Protocols: MCP Server, LangChain/LangGraph, Prompt Engineering, and Multimodal AI.

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Orchestration & Semantics: Camunda for workflow logic, LookML for metrics modeling, and deep familiarity with GCP or Azure cloud native ecosystems.

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Leadership through Architecture: Demonstrated ability to align diverse stakeholders and influence technical direction across organizational boundaries without direct authority.

About Walmart Global Tech

Imagine working in an environment where one line of code can make life easier for hundreds of millions of people. That's what we do at Walmart Global Tech. We're a team of software engineers, data scientists, cybersecurity expert's and service professionals within the world's leading retailer who make an epic impact and are at the forefront of the next retail disruption. People are why we innovate, and people power our innovations. We are people-led and tech-empowered. We train our team in the skillsets of the future and bring in experts like you to help us grow. We have roles for those chasing their first opportunity as well as those looking for the opportunity that will define their career. Here, you can kickstart a great career in tech, gain new skills and experience for virtually every industry, or leverage your expertise to innovate at scale, impact millions and reimagine the future of retail.

Flexible, hybrid work:

We use a hybrid way of working that is primarily in office coupled with virtual when not onsite. Our campuses serve as a hub to enhance collaboration, bring us together for purpose and deliver on business needs. This approach helps us make quicker decisions, remove location barriers across our global team and be more flexible in our personal lives.

Benefits:

Benefits: Beyond our great compensation package, you can receive incentive awards for your performance. Other great perks include 401(k) match, stock purchase plan, paid maternity and parental leave, PTO, multiple health plans, and much more.

Equal Opportunity Employer:

Walmart, Inc. is an Equal Opportunity Employer – By Choice. We believe we are best equipped to help our associates, customers and the communities we serve live better when we really know them. That means understanding, respecting and valuing diversity- unique styles, experiences, identities, ideas and opinions – while being inclusive of all people.

The above information has been designed to indicate the general nature and level of work performed in the role. It is not designed to contain or be interpreted as a comprehensive inventory of all responsibilities and qualifications required of employees assigned to this job. The full Job Description can be made available as part of the hiring process.

At Sam's Club, we offer competitive pay as well as performance-based bonus awards and other great benefits for a happier mind, body, and wallet!

-Health benefits include medical, vision and dental coverage

-Financial benefits include 401(k), stock purchase and company-paid life insurance

-Paid time off benefits include PTO, parental leave, family care leave, bereavement, jury duty, and voting. You will also receive PTO and/or PPTO that can be used for vacation, sick leave, holidays, or other purposes. The amount you receive depends on your job classification and length of employment. It will meet or exceed the requirements of paid sick leave laws, where applicable.

For information about PTO, see <https://one.walmart.com/notices>.

- Other benefits include short-term and long-term disability, company discounts, Military Leave Pay, adoption and surrogacy expense reimbursement, and more.

Live Better U is a company paid education benefit program for full-time and part-time associates in Walmart and Sam's Club facilities. Programs range from high school completion to bachelor's degrees, including English Language Learning and short-form certificates. Tuition, books, and fees are completely paid for by Walmart.

Eligibility requirements apply to some benefits and may depend on your job classification and length of employment. Benefits are subject to change and may be subject to a specific plan or program terms.

For information about benefits and eligibility, see [One.Walmart](#).

The annual salary range for this position is \$143,000.00 - \$286,000.00

Additional compensation includes annual or quarterly performance bonuses.

Additional compensation for certain positions may also include Regional Pay Zone (RPZ) (based on location).

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Minimum Qualifications...

Outlined below are the required minimum qualifications for this position. If none are listed, there are no minimum qualifications.

Option 1: Bachelor's degree in Computer Science or related field and 5 years' experience in data engineering, solution architecture, business intelligence, business analytics or related field. Option 2: 7 years' experience in data engineering, solution architecture, business intelligence, business analytics or related field. Option 3: Master's degree in Computer Science and 3 years' experience in data engineering, solution architecture, business intelligence, business analytics or related field.

Preferred Qualifications...

Outlined below are the optional preferred qualifications for this position. If none are listed, there are no preferred qualifications.

Data architecture/software architecture/data modeling, Master's degree in Computer Science or related field and 5 years' experience in software engineering or related field, Relevant industry experience (for example, retail, supply chain, eCommerce, healthcare, etc.), Solution Architecture, We value candidates with a background in creating inclusive digital experiences, demonstrating knowledge in implementing Web Content Accessibility Guidelines (WCAG) 2.2 AA standards, assistive technologies, and integrating digital accessibility seamlessly. The ideal candidate would have knowledge of accessibility best practices and join us as we continue to create accessible products and services following Walmart's accessibility standards and guidelines for supporting an inclusive culture.

Primary Location...

809 11th Ave, Sunnyvale, CA 94089-4731, United States of America

Walmart and its subsidiaries are committed to maintaining a drug-free workplace and has a no tolerance policy regarding the use of illegal drugs and alcohol on the job. This policy applies to all employees and aims to create a safe and productive work environment.

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 Sunnyvale, CA

 Full time

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About Us



About Walmart:

Fifty years ago, Sam Walton started a single mom-and-pop shop and transformed it into the world's biggest retailer. Since those founding days, one thing has remained consistent: our commitment to helping our customers save money so they can live better. Today, we're reinventing the shopping experience and our associates are at

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